REIM: Recovery-Enhanced Imitation Learning for Failure-Aware Manipulation in Meta-World

Anonymous Authors
Submission under review

Abstract—**Imitation-learned manipulation policies can be accurate on demonstrations yet brittle after a missed grasp, object drop, or control perturbation shifts the closed-loop state distribution. We introduce Recovery-Enhanced Imitation Learning (REIM), a failure-aware controller that combines a state-based Action-Chunking-with-Transformers (ACT) task policy, a causal LSTM risk monitor, and a trigger-aligned supervised recovery option. Its curriculum executes ACT online under disturbances, serializes the complete MuJoCo simulator state at an early collection trigger, and queries a scripted expert for successful continuations from those learner-induced states. A deterministic recovery MLP is trained with Smooth L1 loss on 42,386 state–action pairs and selected on 8,212 disjoint validation pairs. About 99% of the captured states precede any lift, so REIM is principally an early pre-grasp correction system rather than a post-drop repair policy. Across 1,000 paired Meta-World PickPlace trials at 20% mixed simulated disturbance, REIM reaches 90.4% task success versus 73.4% for ACT and 81.3% for one fresh-task retry. The same recovery actor with a semantic heuristic gate reaches 87.7%, while REIM improves success by +2.7 pp and uses 10.4% fewer interventions. At 40% disturbance, REIM retains 63.5% success versus 31.5% for ACT. We additionally define a shared task-conditioned instantiation for the goal-observable, known-task Meta-World MT10 and MT50 suites, with official-clean evaluation separated from a task-universal disturbance extension. All experiments in this paper use the simulated Sawyer model in Meta-World/MuJoCo; no physical-robot evaluation is claimed.**

I. Introduction

Imitation learning offers a direct route from expert demonstrations to robot control, but supervised action error does not capture the distribution induced by the learned policy. Small errors alter future observations and compound over the horizon, a problem formalized by interactive imitation-learning work such as DAgger [1]. Action chunking reduces the effective horizon and has enabled precise manipulation [2], but a chunked policy can still enter states absent from the successful demonstrations. In PickPlace, typical departures include a failed enclosure, premature gripper opening, object displacement, and trajectories that cease to make goal progress.

A useful recovery system must answer three coupled questions: when nominal execution has become risky, which controller should act next, and which states should train that controller. Training a recovery policy from hand-designed "failure-like" resets leaves a train–deploy gap: the reset may match object position while omitting robot velocity, gripper dynamics, controller memory, or task bookkeeping. Conversely, corrective demonstrations collected only from nominal states do not teach a policy how to respond at the learner-induced states where intervention is actually required.

REIM addresses these issues with a closed-loop composition of imitation, runtime risk monitoring, and targeted recovery imitation. Its main contributions are:

- a causal risk-gated controller coupling temporally ensembled ACT, an LSTM risk monitor, and a persistent deterministic recovery option;
- an online trigger-state curriculum that serializes and restores full numeric Meta-World/MuJoCo states observed when disturbed ACT first crosses a broad risk-monitor collection threshold;
- trigger-aligned supervised recovery trained on successful expert continuations with a trajectory-disjoint validation split, exported as a standalone actor with zero reinforcement-learning interactions; and
- a two-level reproducible evaluation design: a PickPlace mechanism study separating classifier, recovery, success, and intervention evidence, plus a publication-gated MT10/MT50 known-task breadth protocol that keeps official clean scores distinct from task-universal disturbed evaluation.

II. Related Work

A. Imitation Learning Under Distribution Shift

Behavior cloning is vulnerable to covariate shift because its training states come from the expert while its test states are induced by the learner. Interactive aggregation methods query corrective actions on learner-induced states [1]. ACT instead predicts temporally coherent action chunks with a conditional variational Transformer and ensembles overlapping predictions [2]. REIM retains ACT as a strong nominal policy and adds an explicit recovery route for states that successful demonstrations do not cover.

B. Runtime Failure Monitoring

Runtime monitors can use predictive uncertainty, action consistency, semantic progress, or supervised failure labels. Recent work separates erratic actions from stalled task progress [3]. REIM uses a lightweight causal LSTM [4] over semantic state history. It estimates the risk that a labeled event is present now or occurs within a short horizon; it never consumes future observations at deployment.

C. Recovery Policies

Recovery RL separates a task policy from a safety-oriented recovery policy [5]. Residual RL adds learned corrective control to a nominal controller [6], while MetaReSkill [7] allocates training to manipulation failure modes and RecoveryChaining [8] learns local policies that reconnect to nominal skills. REIM differs in the source of its recovery initial-state distribution: each start is a complete simulator state produced by the ACT–detector pair at its first online collection trigger. The collection gate is deliberately broader than the frozen deployment gate. This permits targeted corrective imitation from learner-induced states without requiring on-policy recovery interactions. State restoration is used only to construct a controlled simulation dataset; we do not claim that the same reset mechanism is directly available on hardware.

III. Method

A. Problem Formulation

We consider a finite-horizon manipulation MDP with semantic state $s_t \in \mathbb{R}^{21}$ and Cartesian action $a_t \in [-1, 1]^4$. The state concatenates the simulated Sawyer joint positions, end-effector position and quaternion, object position, goal position, and gripper state. The action contains end-effector displacement $(\Delta x, \Delta y, \Delta z)$ and gripper command g. Task success is the environment's explicit object-at-goal event. REIM learns a nominal policy π_{ACT}, risk predictor f_ϕ, and recovery policy π_{rec}. This semantic-state formulation is the PickPlace mechanism instantiation; the shared multi-task representation is specified below rather than silently conflated with the 21-dimensional input.

B. Chunked Nominal Policy

The state-based ACT model is a conditional variational Transformer that predicts a chunk of 20 future actions,

$$\hat{a}_{t:t+20-1} = \pi_{\text{ACT}}(s_t, z), \qquad z = 0 \text{ at inference.} \tag{1}$$

Training minimizes masked L1 reconstruction plus a KL term for the latent posterior. Overlapping predictions for the current action are combined with exponentially decaying temporal weights. The ensemble is cleared whenever control changes between ACT and the recovery option so an unexecuted pre-intervention chunk cannot re-enter the controller after recovery.

C. Causal Risk Monitoring

The detector consumes only the most recent 10 post-transition states. For short prefixes, valid states are placed first and the remainder is zero padded; packed sequence lengths prevent padding from changing the LSTM state. The failure probability is

$$p_t = \sigma\left(g_\phi\left(\text{LSTM}_\phi(s_{t-h_t+1:t})\right)\right), \quad h_t \le 10. \tag{2}$$

Targets indicate whether an online failure event occurs in the inclusive interval $[t, t+10]$. Thus an event observed at the current step is also positive; causal inputs do not make this a pure future-event forecast. Binary cross entropy trains the model on disturbed ACT rollouts, and trajectories rather than individual windows are assigned to disjoint training and validation splits.

D. Online Trigger-State Recovery Curriculum

Approximate failure resets can reproduce an object's displacement without reproducing the dynamic state that caused it. We instead roll out ACT with the trained detector under mixed action, observation, and object disturbances. At the first $p_t \ge 0.10$, before any recovery action is executed, the collector stores the complete numeric simulator snapshot: MuJoCo positions and velocities, mocap targets, actuator/control state, task goal and reset-time geometry, previous observation, step count, and REIM episode bookkeeping. Training and model-selection validation use non-overlapping episode-seed banks. This collection threshold is intentionally lower than the frozen end-to-end gate $\tau_{\text{on}} = 0.20$: the former provides broad risk-triggered recovery starts, whereas the latter defines the deployed intervention operating point. We therefore call the states online-triggered, not deployment-distribution-matched.

A scripted PickPlace expert then continues from the captured state. Every snapshot is retained for provenance. Only expert-successful trajectories contribute recovery-imitation targets. This yields 978 eligible training starts with 42,386 supervised action pairs and 192 model-selection starts with 8,212 pairs. Only 1.12% of training snapshots and 1.04% of validation snapshots record any prior lift. Thus approximately 99% are pre-grasp or early approach corrections, not post-drop recovery states.

E. Trigger-Aligned Recovery Imitation

The recovery option is a deterministic MLP $\pi_{\text{rec}} : \mathbb{R}^{21} \to [-1, 1]^4$ with two 256-unit `tanh` hidden layers and a clipped linear action head. Let $\mathcal{D}_{\text{rec}} = \{(s_i, a_i^E)\}$ contain states and scripted-expert actions from successful continuations after online triggers. We optimize

$$\mathcal{L}_{\text{rec}}(\psi) = \frac{1}{|\mathcal{D}_{\text{rec}}|} \sum_{(s,a^E) \in \mathcal{D}_{\text{rec}}} \text{SmoothL1}\left(\pi_{\text{rec},\psi}(s), a^E\right). \tag{3}$$

This is failure-aware, Smooth-L1 behavior cloning: failure awareness enters through the online LSTM-triggered sampling distribution rather than a hand-labeled mode bit appended to the actor input. Training uses 42,386 pairs, 40 epochs, batch size 512, learning rate 10^{-3}, and Gaussian observation augmentation with standard deviation 0.005. The epoch with minimum Smooth L1 on 8,212 trajectory-disjoint validation pairs is retained. Its validation loss is 0.006391.

The deployed file contains only the 21–256–256–4 actor, observation and action bounds, and provenance metadata. It contains no critic, stochastic log-standard-deviation, optimizer, or rollout state and records zero reinforcement-learning environment interactions and zero policy-gradient updates. An export audit compares the standalone actor against its source on 42,386 training states, 8,212 validation states, and 10,000 deterministic random-support states: all 60,598 clipped and unclipped action outputs are bitwise identical.

F. Closed-Loop Arbitration

At each step the controller updates the causal history and computes p_t. ACT acts while $p_t < \tau_{\text{on}}$; crossing τ_{on} activates the recovery option. The frozen values are $(\tau_{\text{on}}, \tau_{\text{off}}, m, B, c) = (0.20, 0, 150, 150, 200)$, where m is the minimum commitment, B the budget, and c the number of clear observations required for release. Because $m = B$ and $c > B$, the final protocol disables mid-intervention risk-clear release: π_{rec} retains control until task success or budget exhaustion. If the task remains active after an exhausted intervention, ACT resumes with its temporal ensemble reset. All evaluation entry points use this same configuration.

G. Shared Multi-Task Instantiation

For the breadth study, one task-conditioned controller is learned per benchmark suite rather than one controller per task. Given the official raw observation $o_t \in \mathbb{R}^{39}$ and known task index j, the shared input is $x_t = [o_t; e_j]$, where e_j is the official ordered task one-hot. This gives 49 inputs for MT10 and 89 for MT50; the Cartesian action remains four-dimensional. The shared ACT, causal LSTM, and recovery MLP all receive the same conditioned input.

Multi-task risk supervision is deliberately task agnostic in meaning but is not identical to the PickPlace event ontology. During data generation only, the matching Meta-World scripted expert supplies a counterfactual action. For each task k, the behavioral-deviation threshold τ_k is fitted exclusively on that task's failure-training bank as the $q = 0.90$ quantile of ACT–expert action L1 disagreement. The event label is the threshold crossing, OR the final 25 steps of an unsuccessful episode; causal targets then ask whether an event occurs in the inclusive window $[t, t+10]$. All τ_k values are frozen after this training-bank fit and reused unchanged for validation and final evaluation. A fixed raw-disagreement cutoff is retained only as a scale diagnostic, not as a universal training-label threshold, and the deployed LSTM never accesses expert actions. At the first detector trigger, the expert takes over in the same online rollout, and only successful continuations contribute recovery-imitation targets. Thus the multi-task study preserves learner-induced, trigger-aligned correction while making no claim that it uses the exact simulator-snapshot restoration of the PickPlace curriculum.

IV. Experiments

A. Simulation Environment and Data

The mechanism study uses the Meta-World 3.1.1 `pick-place-v3` single-task, goal-observable environment [9] with MuJoCo dynamics [10]. The requested benchmark is the historical PickPlace-v2 Sawyer task. Meta-World 3.1.1 exposes its maintained Gymnasium successor as `pick-place-v3`; the wrapper records both identifiers and maps a v2 request to that maintained registration while preserving the Sawyer pick–move–place task semantics. We report the exact executed identifier to avoid implying use of an unavailable legacy API. The 21-dimensional policy input includes privileged simulator quantities rather than images or a physical sensing stack.

The ACT dataset contains 500 successful scripted trajectories and 25,969 transitions. Failure-predictor data come from 2,000 disturbed ACT episodes, using Gaussian action and observation noise plus one-shot simulated object-position displacements. Recovery starts are collected at 20% nominal disturbance scaling. The training and validation banks contain 985 and 193 exact trigger snapshots, respectively, with disjoint seed ranges and checkpoint hashes recorded in their manifests.

B. Training Protocol

ACT uses hidden width 256, 8 attention heads, 3 encoder layers, 4 decoder layers, and a 32-dimensional latent. It is trained for 200 epochs. The detector has an LSTM width of 128 followed by an 64-dimensional MLP. The recovery actor uses two 256-unit `tanh` layers and is optimized for 40 supervised epochs with batch size 512, learning rate 10^{-3}, and input-noise standard deviation 0.005. Training and validation contain 42,386 and 8,212 trigger-aligned pairs, respectively. Recovery-policy training uses no environment interaction; the primary uncertainty estimates therefore come from paired held-out episodes, not multiple policy-gradient seeds.

C. Baselines and Ablations

The paired baseline comparison contains: *ACT*, the nominal policy without intervention; *ACT + Random Reset*, one fresh-task retry after a heuristic failure event (the retry uses a new simulator seed); *ACT + Heuristic Recovery*, the same trigger-aligned imitation actor activated by a semantic heuristic rather than the LSTM; and *REIM*, which uses the LSTM gate with that actor. The heuristic fires after at least 20 steps and observed object interaction when a 12-step goal-distance window improves by less than 10^{-3} or worsens by more than 5×10^{-3}; explicit drop/failure flags also fire it. It therefore uses simulator-level semantic information. The paper-facing component study uses ACT, heuristic-gated recovery, and full REIM. The first removes recovery entirely; the second replaces the learned gate with simulator-level semantics while retaining the identical recovery actor.

D. Evaluation and Metrics

The PickPlace main evaluation uses 1,000 paired CRN episode specifications per method under 20% mixed disturbance, spanning seeds 8,000,042–8,001,041. Robustness uses 200 paired episodes at each of 0%, 10%, 20%, 30%, and 40%. Each episode records explicit task success, intervention count, task completion while the recovery controller is active, and length. For learned recovery methods, "recovery rate" is the number of task completions reached during active recovery divided by the number of recovery interventions. Random Reset instead reports post-reset completion per reset and is labeled separately; the two intervention outcomes are not compared as identical quantities. Neither is a detector-risk-clearance proxy. Success intervals are Wilson intervals; ratio and step-count intervals are obtained by episode bootstrap.

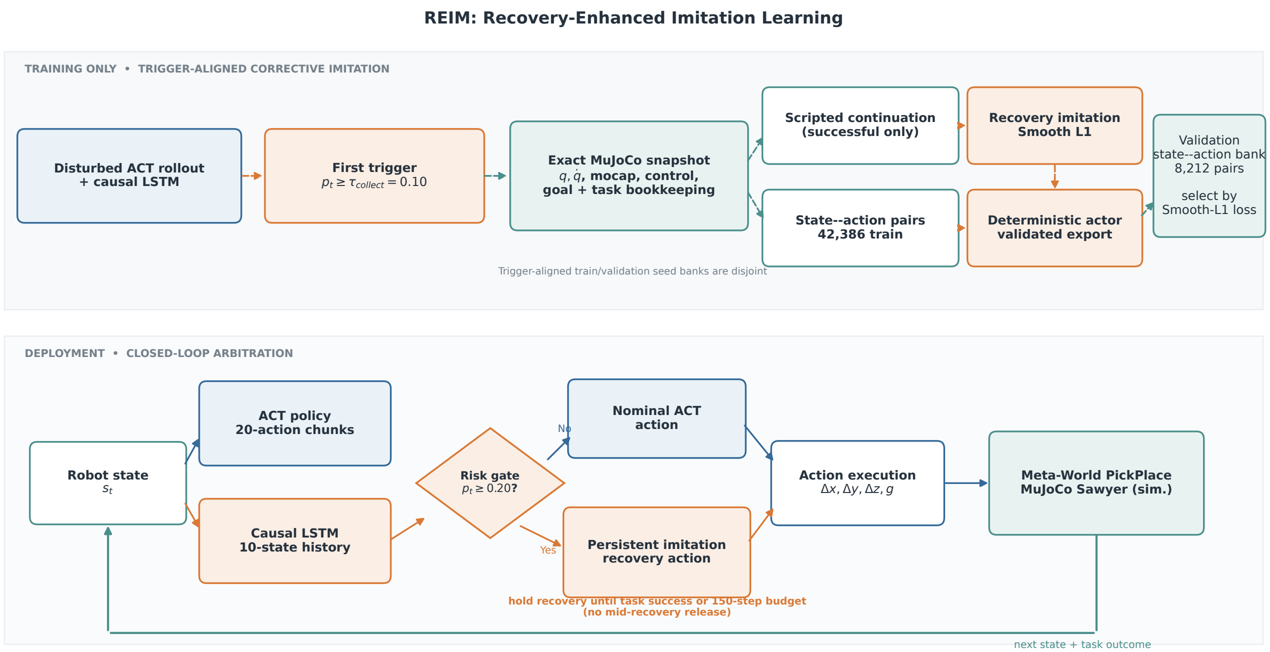

Fig. 1. REIM training and deployment architecture. The environment block is explicitly the Meta-World PickPlace task running in MuJoCo with a simulated Sawyer model. Dashed training-only paths denote exact simulator snapshot capture, scripted corrective demonstrations, and validation-loss checkpoint selection; the scripted expert is never queried during evaluation.

Every nominal attempt in a condition consumes the same serialized common-random-number (CRN) episode specification: task identity, initial MuJoCo state, disturbance schedule, and episode seed are hash-checked before aggregation. The Random Reset baseline alone receives one explicitly labeled fresh-task retry whose second-attempt specification uses a reserved, deterministically offset seed. The $\tau_{\text{on}} = 0.20$ LSTM gate is frozen before the primary evaluation. A separate post-freeze sweep on 200 previously unused CRN episodes is reported only as a sensitivity diagnostic; it does not select or alter the primary gate.

E. Known-Task MT10/MT50 Protocol

The breadth protocol follows Meta-World's goal-observable multi-task setting: MT10 and MT50 contain known task families, and the ordered task identity is supplied to every method. It is therefore a shared known-task evaluation, not an unseen-task, transfer, or meta-learning claim. Each task contributes the same number of successful expert demonstrations and 50 final episodes. Clean evaluation uses the official environment success signal and a 500-step horizon. Separate task-bank seeds and payload hashes isolate demonstration, failure, recovery, validation, and final-evaluation data.

The comparison contains a shared two-layer MLP behavior-cloning capacity baseline, shared ACT, ACT with the learned recovery actor activated by a semantic heuristic, and full REIM. The primary statistic is task-macro success; we also retain per-task success, worst-quartile task success, intervention burden, and a task-stratified paired bootstrap against ACT.

The zero-disturbance condition is the official-clean anchor. Nonzero levels in 0%, 10%, 20%, 30%, and 40% apply only task-universal action and observation noise, never a PickPlace-specific object teleportation, and are labeled a REIM robustness extension rather than official Meta-World scores. A fresh-task retry is appendix-only and does not enter the primary multi-task comparison.

V. Results

A. PickPlace Mechanism Results

REIM achieves 90.4% task success, compared with 73.4% for ACT, 81.3% for random reset, and 87.7% for semantic-heuristic recovery. Across REIM interventions, 85.3% reach task completion while recovery retains control. Pairing gives a REIM–ACT gain of +17.0 pp (95% CI [+14.7, +19.3] pp) and a REIM–Random Reset gain of +9.1 pp (CI [+6.1, +12.1] pp). Against the stronger semantic-heuristic gate, REIM improves success by +2.7 pp (CI [+0.7, +4.8] pp) while intervening in 65.5% rather than 73.1% of episodes (655 versus 731 of 1,000). This corresponds to 76 fewer intervened episodes and a 10.4% relative burden reduction. Thus the frozen LSTM gate Pareto-dominates this simulator-privileged heuristic in the primary CRN benchmark; the paired gate audit below tests whether this persists at matched burden.

From no disturbance to the strongest condition, REIM changes from 100.0% to 63.5%, while ACT changes from 100.0% to 31.5%. At 10%, 20%, 30%, and 40% disturbance, the paired success improvements are +2.5 pp, +15.5 pp, +26.0 pp, and +32.0 pp, respectively.

TABLE I

Paired comparison on the simulated Meta-World `pick-place-v3` task. Every method uses the same 1,000 seeds. Success and intervention outcomes are percentages with 95% intervals. For Heuristic Recovery/REIM, recovery is task completion while recovery has control, divided by intervention count; Random Reset reports post-reset completion per reset and is not directly comparable.

Method	Task success ↑	Intervention outcome ↑	Intervention rate	Steps ↓
ACT	73.4% [70.6, 76.0]%	–	–	93.3
ACT + Random Reset	81.3% [78.8, 83.6]%	74.5% [71.3, 77.6]%	73.2%	93.3
ACT + Heuristic Recovery	87.7% [85.5, 89.6]%	83.3% [80.6, 85.9]%	73.1%	72.8
REIM (ours)	90.4% [88.4, 92.1]%	85.3% [82.5, 88.0]%	65.5%	69.3

Paired REIM gain over ACT: +17.0 pp (95% CI [+14.7, +19.3] pp); 170 ACT failures are rescued and 0 ACT successes are harmed. REIM intervenes in 76 fewer episodes than heuristic-gated recovery (10.4% relative reduction).

Figure 3 compares gate operating points on the separate diagnostic bank. At the frozen $\tau_{\text{on}} = 0.20$ point, REIM reaches 90.5% success with 63.5% intervention, +4.5 pp above the matched heuristic (CI [+0.5, +8.5] pp, $p = 0.0490$, exact two-sided McNemar/binomial test). At $\tau_{\text{on}} = 0.175$, its intervention rate 76.5% nearly matches the heuristic's 76.0%: the paired difference is +0.5 pp (CI [-8.0, +8.5] pp). REIM nevertheless reaches 92.5% versus 86.0%, a paired gain of +6.5 pp (CI [+2.5, +11.0] pp, $p = 0.0044$). This post-freeze burden-matched comparison supports the learned gate's utility but is not substituted for the primary benchmark or used for threshold selection.

B. *Known-Task Multi-Task Breadth*

Table II reports the official-clean known-task breadth evaluation. Each task-macro point estimate is accompanied by a 95% within-task episode-bootstrap interval. The paired MT-REIM–MT-ACT task-macro differences are +0.4 pp (95% CI [+0.0, +1.0] pp) on MT10 and +0.0 pp (CI [-0.4, +0.6] pp) on MT50. Nonzero-disturbance results are reported separately as a robustness extension and are not described as official clean benchmark scores.

C. *Completed Component Diagnostics*

ACT reaches validation action L1 0.004013 on a trajectory-grouped split. At the separately reported 0.20 detector operating point, precision is 91.8% and recall is 92.3% over 56,253 validation windows. Figure 5 shows the corresponding confusion matrix. The diagnostic is evaluated at the same frozen probability threshold as end-to-end arbitration, $\tau_{\text{on}} = 0.20$, but window-level classification metrics do not by themselves establish closed-loop utility.

The inclusive target requires a second, stricter audit. Of the positive validation windows, 86.1% have event offset zero. After excluding those current-event windows and scoring only future offsets 1–10 against windows with no event in the horizon, precision/recall/F1 are 53.5%/ 68.3%/60.0%. At trajectory level, the monitor alerts strictly before the first event in 191/258 (74.0%) event trajectories, with median latest alert lead 1 step. We therefore describe the LSTM as a causal risk monitor, not a calibrated long-horizon forecaster.

The recovery actor reaches validation Smooth L1 0.006391. The standalone checkpoint records zero environment-training steps and contains only 72,452 actor parameters. Its export audit finds exactly zero action error on 60,598 states, including every training and validation state plus 10,000 random-support states. This establishes deployment-artifact equivalence; held-out paired episodes, rather than the export check, support the task-level claims.

VI. Recovery and Gate Ablations

Removing recovery entirely reduces the controller to ACT and 73.4% success. Keeping the same trigger-aligned recovery actor but replacing the LSTM with simulator-semantic heuristics reaches 87.7% and intervenes in 73.1% of episodes. Full REIM reaches 90.4% while intervening in only 65.5%, improving both success and intervention burden. The separate burden-matched audit in Fig. 3 makes the gate comparison under nearly equal intervention rates.

Table IV therefore supports two scoped conclusions: recovery is responsible for the large improvement over nominal ACT, and learned temporal arbitration improves on a strong simulator-privileged gate. It does not isolate online snapshot collection from recovery-imitation training, so we make no separate marginal claim about those coupled design choices.

VII. Limitations

The current study has several boundaries. First, its mechanism evidence and component ablations concentrate on state-observed PickPlace. The MT10/MT50 breadth protocol supplies known task identity and therefore cannot establish unseen-task generalization, meta-learning, or transfer; visual perception and contact sensing also remain outside scope. Second, exact simulator restoration in the PickPlace curriculum depends on privileged MuJoCo state and task bookkeeping. It is a controlled way to align recovery training in simulation, not a deployable physical-robot reset mechanism. Hardware would require state estimation and safe physical repositioning or real failure collection. Third, filtering recovery-imitation trajectories by scripted-expert success biases training toward recoverable states and may exclude precisely the hardest failures. Fourth, detector metrics are measured on rollouts from a fixed nominal policy in each protocol and may shift as ACT or the disturbance distribution changes. Moreover, the PickPlace inclusive label makes the standard confusion matrix predominantly a current-event diagnostic, and 66.2% of validation trajectories without a labeled event still contain an alert at the deployed threshold. Fifth, the main table is

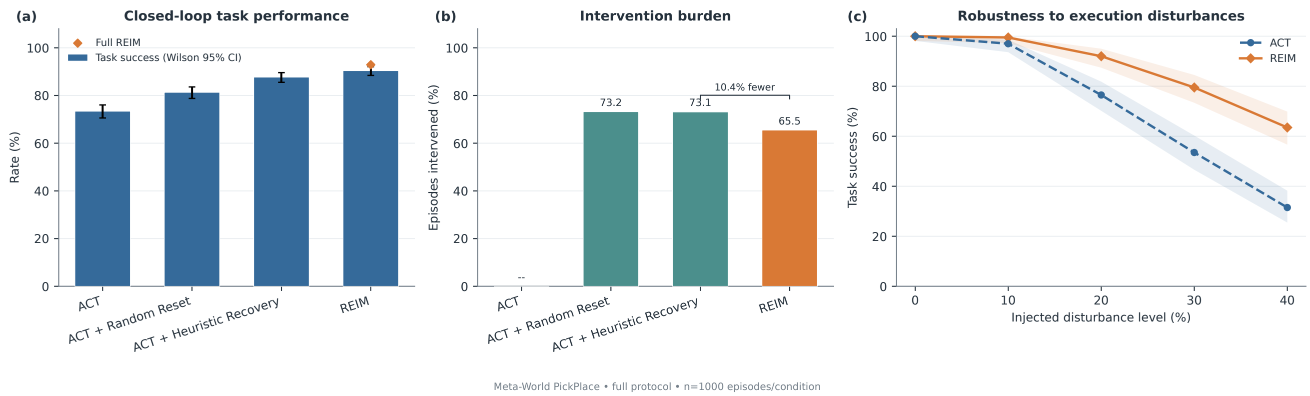

Fig. 2. Final simulated Meta-World results. Left: paired task success for the four methods. Center: the fraction of episodes receiving an intervention, exposing controller burden when success rates are similar. Right: success under increasing mixed action and observation noise plus simulated one-shot object-position displacements. Every panel is regenerated from final per-episode records; no development-run curve is substituted.

TABLE II

OFFICIAL-CLEAN, GOAL-OBSERVABLE KNOWN-TASK EVALUATION. ALL METHODS RECEIVE THE SAME OFFICIAL TASK ONE-HOT, TASK PAYLOADS, AND PAIRED EPISODE SEEDS. SUCCESS IS AVERAGED EQUALLY ACROSS TASKS; BRACKETS ARE 95% WITHIN-TASK EPISODE-BOOTSTRAP INTERVALS. DIFFERENCES USE A PAIRED, TASK-STRATIFIED BOOTSTRAP AGAINST MT-ACT. WORST QUARTILE IS THE MEAN OF THE LOWEST $\lceil T/4 \rceil$ TASK RATES, AND INTERVENTION IS THE TASK-MACRO FRACTION OF EPISODES IN WHICH RECOVERY ACTS. NONZERO DISTURBANCES ARE EXCLUDED AND REPORTED ONLY AS A NON-OFFICIAL REIM ROBUSTNESS EXTENSION.

Benchmark	Method	Task macro [95% CI] ↑	Δ vs. MT-ACT [95% CI]	Worst quartile ↑	Interv.
MT10	MT-MLP BC	91.4% [89.6, 93.2]%	-5.2 pp [-7.4, -3.0] pp	72.0%	–
	MT-ACT	96.6% [95.2, 98.0]%	–	88.7%	–
	Heuristic-gated recovery	96.6% [95.2, 98.0]%	+0.0 pp [-1.6, +1.6] pp	88.7%	22.2%
	MT-REIM (ours)	**97.0% [95.6, 98.2]%**	**+0.4 pp [+0.0, +1.0] pp**	**90.0%**	46.6%
MT50	MT-MLP BC	81.3% [80.2, 82.4]%	-10.8 pp [-12.0, -9.5] pp	40.8%	–
	MT-ACT	92.1% [91.3, 92.9]%	–	71.2%	–
	Heuristic-gated recovery	94.8% [94.0, 95.5]%	+2.7 pp [+2.0, +3.4] pp	80.8%	17.3%
	MT-REIM (ours)	**92.2% [91.3, 93.0]%**	**+0.0 pp [-0.4, +0.6] pp**	**71.4%**	28.4%

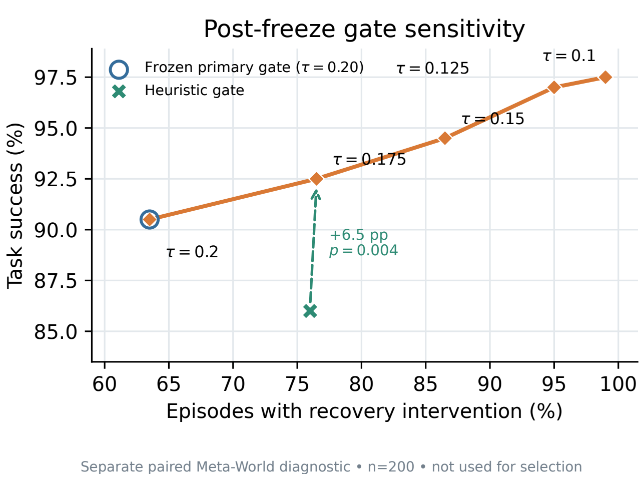

Fig. 3. Post-freeze gate sensitivity on a separate set of 200 paired Meta-World seeds at 20% disturbance. Each point is labeled by τ_{on}; the green cross is the semantic heuristic evaluated on the same serialized CRN bank. The circled 0.20 operating point remains the frozen primary gate, while the dashed comparison at 0.175 exposes matched intervention burden. This diagnostic was not used for model or threshold selection.

TABLE III

COMPLETED COMPONENT DIAGNOSTICS. THESE VALUES USE DIFFERENT VALIDATION UNITS AND ARE NOT END-TO-END COMPARISONS. THE RECOVERY ROW USES TRAJECTORY-DISJOINT EXACT-TRIGGER VALIDATION PAIRS AND IS THEREFORE MODEL-SELECTION EVIDENCE RATHER THAN AN UNBIASED TEST RESULT.

Component	Diagnostic	Value
ACT	validation action L1	0.004013
LSTM	precision at $\tau = 0.20$	91.8%
LSTM	recall at $\tau = 0.20$	92.3%
Recovery imitation	validation Smooth L1	0.006391
Standalone actor	audited output error	exactly zero

TABLE IV

RECOVERY AND GATE ABLATION UNDER THE PAIRED PROTOCOL OF TABLE I. THE HEURISTIC AND LSTM ROWS USE THE SAME TRIGGER-ALIGNED RECOVERY ACTOR, ISOLATING THE ARBITRATION RULE.

Configuration	Success ↑	Intervention
ACT (no recovery)	73.4%	–
Heuristic gate + recovery	87.7%	73.1%
REIM (LSTM gate)	90.4%	65.5%

conditional on one trained ACT, detector, and supervised recovery actor; the multi-task protocol likewise uses one trained stack per suite. Paired episode uncertainty therefore does not propagate all model-training uncertainty. Sixth, the validation pairs select the recovery epoch and therefore provide model-

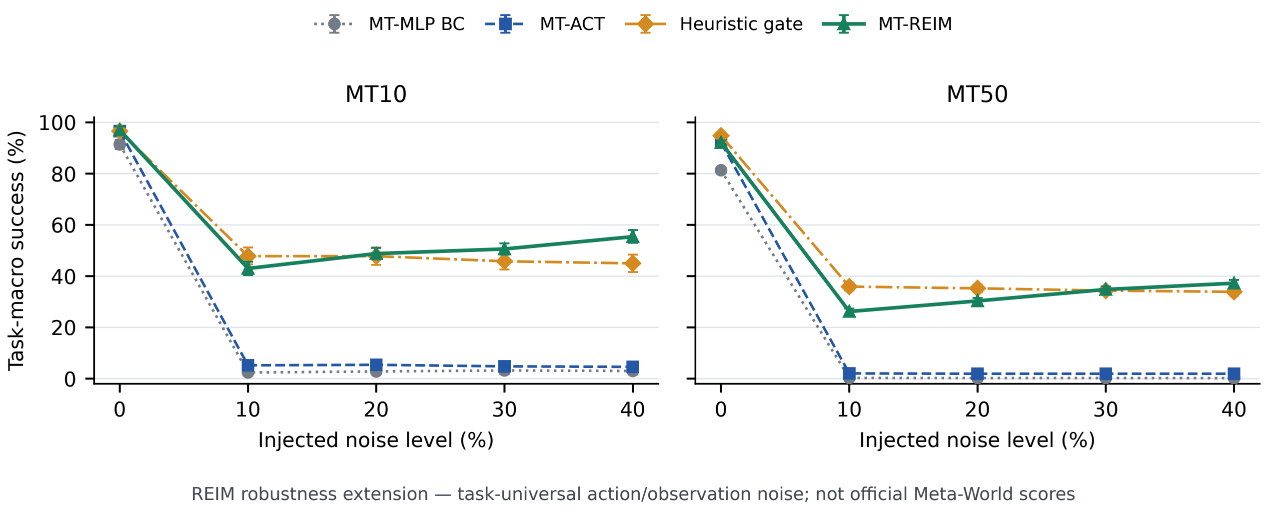

Fig. 4. Non-official REIM robustness extension on the known-task, goal-observable MT10 and MT50 suites. Curves apply task-universal Gaussian action and observation noise; they do not alter object poses. Points are task-macro success over 50 episodes per task, and bars are 95% within-task episode-bootstrap intervals. These disturbed conditions are deliberately separated from the official-clean scores in Table II.

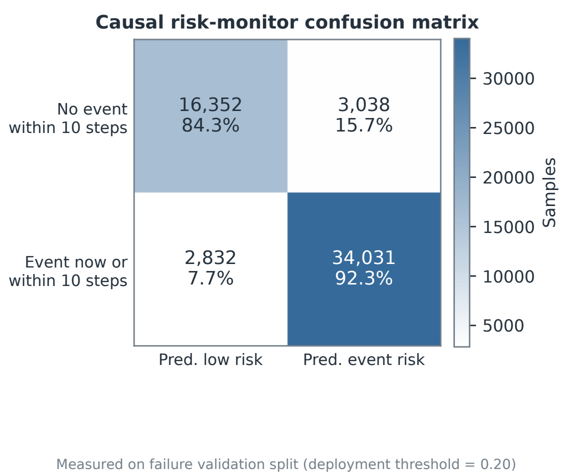

Fig. 5. Causal risk-monitor confusion matrix on the trajectory-grouped validation split at $\tau = 0.20$. Counts are $(\mathrm{TN}, \mathrm{FP}, \mathrm{FN}, \mathrm{TP}) = (16{,}352, 3{,}038, 2{,}832, 34{,}031)$.

selection rather than independent test evidence. Finally, the reported "intervention outcome" is task completion while the recovery option remains in control: the persistent option does not demonstrate recovery to a certified safe set followed by an ACT hand-back. It can also waste steps after false positives, so all arbitration parameters remain frozen throughout the paired benchmark.

VIII. Conclusion

REIM combines state-based chunked imitation, causal runtime risk monitoring, and a persistent trigger-aligned imitation recovery option in one closed-loop manipulation controller. Corrective demonstrations are collected from complete simulator states captured at online risk-monitor triggers, and a deterministic recovery actor is selected by trajectory-disjoint validation loss without reinforcement-learning interaction. In paired PickPlace Meta-World/MuJoCo evaluation, REIM improves ACT by +17.0 pp and remains substantially more robust at strong disturbances. Its LSTM gate reduces interventions by 10.4% relative to heuristic-triggered recovery while improving success by +2.7 pp in the primary paired benchmark; a separate burden-matched audit yields +6.5 pp at $p = 0.0044$. A shared task-conditioned instantiation additionally defines a known-task MT10/MT50 breadth test whose official-clean evidence is kept separate from the task-universal disturbance extension. The captured distribution is approximately 99% pre-grasp/early correction, so the evidence supports selective simulated failure prevention and recovery, not general post-drop repair, visual perception, sim-to-real transfer, or a physical Sawyer claim.

References

[1] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 15. PMLR, 2011, pp. 627–635. [Online]. Available: https://proceedings.mlr.press/v15/ross11a.html

[2] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," in *Proceedings of Robotics: Science and Systems*, Daegu, Republic of Korea, Jul. 2023.

[3] C. Agia, R. Sinha, J. Yang, Z. Cao, R. Antonova, M. Pavone, and J. Bohg, "Unpacking failure modes of generative policies: Runtime monitoring of consistency and progress," in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 689–723. [Online]. Available: https://proceedings.mlr.press/v270/agia25a.html

[4] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[5] B. Thananjeyan, A. Balakrishna, S. Nair, M. Luo, K. Srinivasan, M. Hwang, J. E. Gonzalez, J. Ibarz, C. Finn, and K. Goldberg, "Recovery RL: Safe reinforcement learning with learned recovery zones," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 4915–4922, 2021.

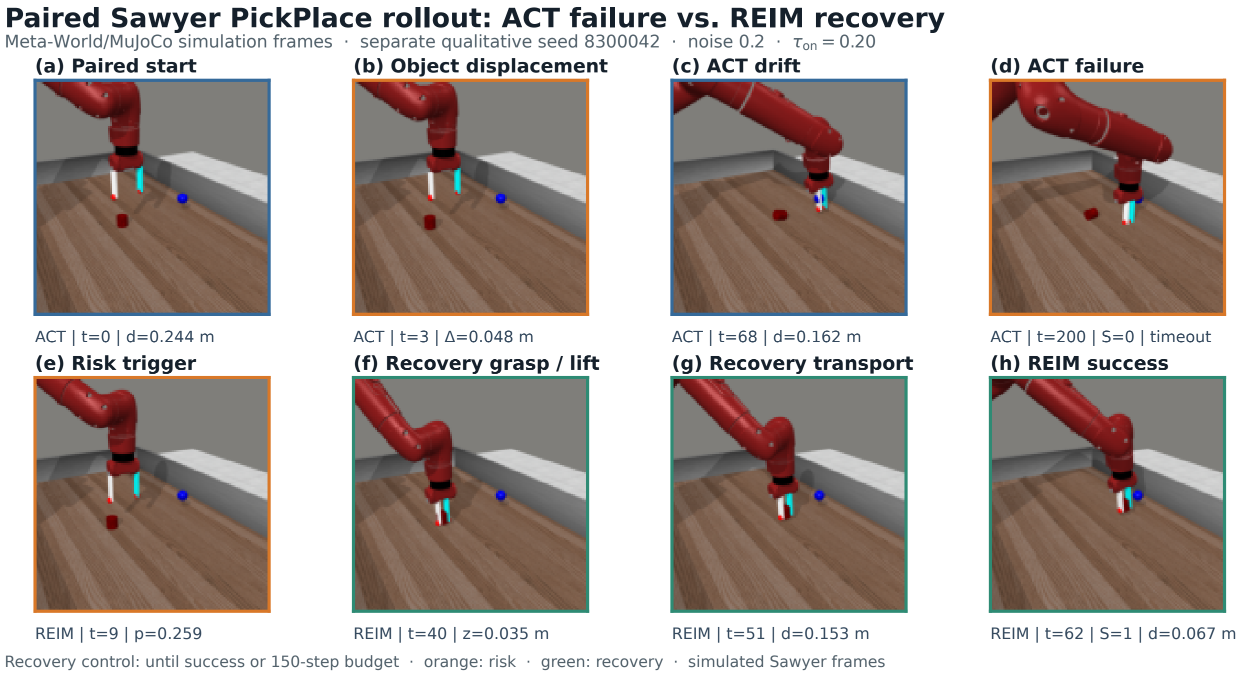

Fig. 6. Paired qualitative operation sequence from separate simulation seed 8,300,042 at 20.0% disturbance. With the frozen evaluation gate $\tau_{\text{on}} = 0.20$, ACT does not complete the task within 200 steps, whereas REIM triggers once and completes in 62 steps, including 53 supervised-recovery actions. These are rendered Meta-World/MuJoCo frames of the simulated Sawyer model, not hardware photographs. This selected trace explains controller operation and does not estimate aggregate success.

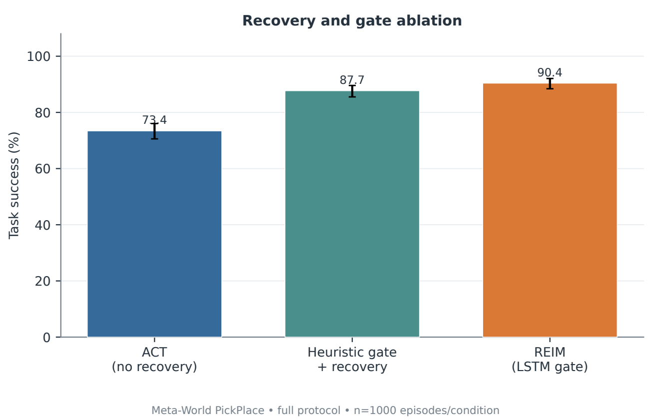

Fig. 7. Paper-facing component removal study. ACT removes recovery; Heuristic gate + recovery replaces only the learned LSTM arbitration rule while retaining the same recovery actor; REIM uses LSTM-triggered recovery imitation. Error bars are Wilson 95% intervals.

[6] T. Johannink, S. Bahl, A. Nair, J. Luo, A. Kumar, M. Loskyll, J. A. Ojea, E. Solowjow, and S. Levine, "Residual reinforcement learning for robot control," in *2019 International Conference on Robotics and Automation*, 2019, pp. 6023–6029.

[7] S. Vats, M. Likhachev, and O. Kroemer, "Efficient recovery learning using model predictive meta-reasoning," in *2023 IEEE International Conference on Robotics and Automation*, 2023, pp. 7258–7264.

[8] S. Vats, D. K. Jha, M. Likhachev, O. Kroemer, and D. Romeres, "Recoverychaining: Learning local recovery policies for robust manipulation," *arXiv preprint arXiv:2410.13979*, 2024. [Online]. Available: https://arxiv.org/abs/2410.13979

[9] T. Yu, D. Quillen, Z. He, R. Julian, K. Hausman, C. Finn, and S. Levine, "Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning," in *Proceedings of the Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 100. PMLR, 2020, pp. 1094–1100. [Online]. Available: https://proceedings.mlr.press/v100/yu20a.html

[10] E. Todorov, T. Erez, and Y. Tassa, "MuJoCo: A physics engine for model-based control," in *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2012, pp. 5026–5033.